

WENJIN YANG

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I am now a postdoc in the Nanjing University. I mainly work on astronomical masers (e.g. CH₃OH, HCN, SiO, H₂O) in star formation regions, evolved stars and supernova remnants.

Updated on June 25, 2026

RESEARCH INTERESTS

- Astronomical masers
- Star formation
- Circumstellar envelopes of evolved stars
- Supernova remnants interacting with ISM
- Kinematics and dynamics of interstellar medium

EMPLOYMENT

- **Nanjing University** **2023.09–present**
Postdoc (2026.09–present) Collaborate with Dr. Shanghuo Li
Postdoc (2023.09–2026.08) Collaborate with Dr. Ping Zhou
- **Max-Planck-Institut für Radioastronomie** **2020.11–2023.06**
Postdoc Collaborate with Prof. Dr. Karl. M. Menten
Scientific employee of SOFIA/GREAT (2023.03–2023.06)
Guest of Menten's group (2023.07–2025.10)

EDUCATION

- **Purple Mountain Observatory & University of science and technology of China** **2014.09–2020.07**
Ph.D in Astrophysics Advisors: Prof. Dr. Ye Xu, Prof. Dr. Xi Chen
Thesis: *Observational studies of class I methanol masers*
- **Wuhan University** **2010.09–2014.06**
Bachelor of Engineering in Geographical Information System (GIS)

PRESENTATIONS

- 2025.12 [[contributed talk](#)] The 789 key science project meeting: Fundamental Evolution of the Galactic Ecosystem, Nanjing University, Suzhou, China, "Multi-channel feedback of SNRs to ISM: ALMA view of HCO⁺ & CO molecular gas around SNR W49B"
- 2025.09 [[invited talk](#)] Astrophysics seminar, Shanghai Astronomical Observatory, Shanghai, China, "'Rare' masers in our Galaxy"
- 2025.08 [[poster+flash talk](#)] 839. WE-Heraeus-Seminar: Charting the Cosmos: From Cosmic Stellar Nurseries to Evolved Stars using High Powered Telescopes, Ingelheim, Germany, "New sub-millimetre HCN lasers in carbon-rich AGB stars" ([poster_pdf](#), [flash_talk\(3min\)_pdf](#))
- 2025.08 [[contributed talk](#)] Symposium on molecular clouds and star formation 2025, Emeishan, China, "Methanol masers and absorption features in massive star formation regions"
- 2025.07 [[invited talk](#)] Xinjiang Astronomical Observatory, Xinjiang, China, "New sub-millimetre HCN lasers in carbon-rich AGB stars" ([pdf](#))
- 2025.07 [[contributed talk](#)] The 12th Interstellar Physics and Chemistry workshop, Zhangye, China, "Discovery of new sub-millimetre HCN lasers in carbon-rich AGB stars"
- 2024.02 [[contributed talk](#)] The second Cross-Strait Workshop on Radio Astronomy, Shanghai, China, "Methanol masers and absorption features in massive star formation regions" ([pdf](#))
- 2023.10 [[talk](#)] Group meeting of Millimeter and Submillimeter Astronomy in MPIfR, Bonn, Germany,

"Maser Investigation toward Off-Plane Stars: detection of SiO masers in the Galactic thick disk and halo" ([pdf](#))

- 2023.06 [[invited talk](#)] Nanjing University, Nanjing, China, "Masers in star formation regions and evolved stars"
- 2023.03 [[poster+flash talk](#)] IAU 380 Cosmic Masers, Kagoshima, Japan, "ATLASGAL: Methanol masers at 3 mm" ([pdf](#))
- 2022.06 [[poster](#)] The MPIfR's Scientific Advisory Committee (Fachbeirat), Bonn, Germany, "Probing infall in high-mass star-forming regions from red-shifted absorption of CH₃OH and HNCO"
- 2021.11 [[invited talk \(on-line\)](#)] Guangzhou University, China, "How to use RADEX code"
- 2019.07 [[contributed talk](#)] Symposium on molecular clouds and star formation 2019, Altay, China, "44 GHz Methanol Masers: Observations toward 95 GHz Methanol Masers"
- 2017.10 [[contributed talk](#)] Symposium on molecular clouds and star formation 2017, Yichang, China, "The new catalog of 95 GHz methanol maser"
- 2016.11 [[contributed talk](#)] The Chinese Annual Astronomy/Astrophysics Meeting, Wuhan, China, "The Current Status of 95 GHz methanol masers observations"

ACCEPTED PROPOSALS

PI Proposals

- [The Karl G. Jansky Very Large Array \(JVLA\)](#)
(24A-198: 9 h; 24B-109: 3.5 h)
- [The Australia Telescope Compact Array \(ATCA\)](#)
(C3471: 9.5 h)
- [The Atacama Pathfinder EXperiment \(APEX\)](#)
(M9509B_111: 20 h; M9505C_115: 26.5 h)
- [The IRAM-30m telescope \(IRAM-30m\)](#)
(141-22: 29 h, 112-23)
- [The Effelsberg-100 m](#)
(17-21: 22 h, 65-17: 29 h)
- [The Very Long Baseline Array \(VLBA\)](#)
(17A-112: 24 h)
- [The PMO-13.7 m](#)
(20A-007: 147 h, 18A-001: 57 h, 17A-007: 130 h)

Co-I. Proposals (> 1000 hours)

[ALMA](#) (2023.1.01576.S), [VLA](#) (23A-136), [VLBA](#) (20B-107), [ATCA](#) (C3457), [EAVN](#) (EAVN 2024A 378)
[APEX-12 m](#) (M9519A_109, M9505B_113, M9509C_113), [Effelsberg-100 m](#) (92-20, 13-21, 92-21, 95-21, 100-21, 34-22, 75-22, 12-23, 14-23, 111-24, 112-24, 101-25, 112-25), [IRAM-30 m](#) (028-21, 137-24, 014-25, 015-25, 036-25), [KVN](#) (KVN-16B-SD-03), [Yebes-40 m](#) (23A009, 25A024), [ARO-12 m](#) (Gong_25a_1)

PROFESSIONAL SERVICE

2025.11	Host of Dr. Patricia Reich's visit in Nanjing University
2025.08	Chair of Galactic surveys session in 839. WE-Heraeus-Seminar (Charting the Cosmos)
2025.04	Host of Prof. Dr. Mark A. Thompson's visit in Nanjing University
2025.03	LOC of Symposium on diffuse sources for a new era of X-ray observations (Einstein Probe internal symposium)
2024.03 – 2024.07	Coordinator and host of the (weekly) MARTES Talk in School of Astronomy and Space Science, Nanjing University

FUNDING

2025.01 – 2027.12	National Natural Science Foundation of China (12403027; ¥ 300,000) "Exploring the physical environment in star-forming regions and near supernova remnants through multiple maser lines"
2024.09 – 2026.09	China Postdoctoral Science Foundation (2024M751376; ¥ 80,000)

HONOR AND AWARDS

- 2020 Outstanding graduate, University of science and technology of China
- 2017 National scholarship for master student, University of science and technology of China
- 2014-2015 Merit student, University of Chinese Academy of Sciences

SKILLS OF NOTE

AI tool	Cursor, Codex
Software/Language	GILDAS, CASA, MIRIAD (basic), CARTA, python, html/css (basic), markdown
Radiative transfer code	RADEX/myRadex, molpop-cep, Cassis (basic)
Observing experience	Effelsberg-100m (remote), IRAM-30m (remote), APEX-12m (remote), ARO-12m (remote), ATCA (remote), PMO-13.7m (on site)

REFERENCES

Dr. Shanghuo Li (Nanjing University, China)

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Dr. Ping Zhou (Nanjing University, China)

E-mail: pingzhou@nju.edu.cn

Dr. Christian Henkel (Max-Planck-Institut für Radioastronomie, Germany)

E-mail: chenkel@mpifr-bonn.mpg.de

Prof. Dr. Ye Xu (Purple Mountain Observatory, Chinese Academy of Science, China)

E-mail: xuye@pmo.ac.cn

Prof. Dr. Xi Chen (Guangzhou University, China)

E-mail: chenxi@gzhu.edu.cn

Prof. Dr. Karl M. Menten[†] (Max-Planck-Institut für Radioastronomie, Germany)

[†]Prof. Dr. Karl Martin Menten passed away in 2024 at the age of 67. We miss him very much and look back with gratitude on the time we spent together.

REFEREED PUBLICATIONS

A full list via ADS

First authored Publications:

6. [New submillimetre HCN lasers in carbon-rich evolved stars](#)

W. Yang, K. T. Wong, H. Wiesemeyer, K. M. Menten, Y. Gong, J. Cernicharo, E. De Beck, B. Klein, C. A. Durán
2025, *A&A*, 696, A60

This work is dedicated to Karl M. Menten

5. [Maser Investigation toward Off-Plane Stars \(MIOPS\): detection of SiO masers in the Galactic thick disk and halo](#)

Wenjin Yang, Yuanwei Wu, Yan Gong, Nicolas Maunon, Bo Zhang, Karl M. Menten, Xiaofeng Mai, Dejian Liu, Juan Li, Jingjing Li, 2024, *ApJ*, 961, 190

4. [ATLASGAL: 3-mm class I methanol masers in high-mass star formation regions](#)

W. Yang, Y. Gong, K. M. Menten, J. S. Urquhart, C. Henkel, F. Wyrowski, T. Csengeri, S. P. Ellingsen, A. R. Bemis, J. Jang, 2023, *A&A*, 675, A112

3. [Redshifted methanol absorption tracing infall motions of high-mass star formation regions](#)

W. J. Yang, K. M. Menten, A. Y. Yang, F. Wyrowski, Y. Gong, S. P. Ellingsen, C. Henkel, X. Chen, Y. Xu, 2022, *A&A*, 658, A192

2. [44GHz Methanol Masers: Observations toward 95GHz Methanol Masers](#)

Wenjin Yang, Ye Xu, Yoon Kyung Choi, Simon P. Ellingsen, Andrej M. Sobolev, Xi Chen, Jingjing Li, Dengrong Lu, 2020, *ApJS*, 248, 18

1. [A New 95 GHz Methanol Maser Catalog. I. Data](#)

Wenjin Yang, Ye Xu, Xi Chen, Simon P. Ellingsen, Dengrong Lu, Binggang Ju, Yingjie Li, 2017, *ApJS*, 231, 20

Co-authored Publications:

17. [The most luminous extragalactic class I methanol maser reveals bar-driven shocks in NGC 1365](#)

Xi Chen, Tian Yang, Shi-Min Song, Simon Ellingsen, Shari Breen, Tiege McCarthy, **Wenjin Yang**, Ye-Wei Mao, 2026, *ApJL*, 999, L20

16. [Oxygen isotopes reveal low-mass star dominance in the Small Magellanic Cloud](#)

Yan Gong, Zhi-yu Zhang, Christian Henkel, C.-H. Rosie Chen, **Wenjin Yang**, Xindi Tang, Leslie K. Hunt, Axel Weiss, Gang Wu, Yaoting Yan, Konstantin Grishunin, Karl M. Menten, 2025, *ApJ*, 994, 158

15. [Shock-induced HCNH⁺ abundance enhancement in the heart of the starburst galaxy NGC 253 unveiled by ALCHEMI](#)

Y. Gong, C. Henkel, C. T. Bop, J. G. Mangum, E. Behrens, F. J. Du, S. B. Zhang, S. Martin, K. M. Menten, N. Harada, M. Bouvier, X. D. Tang, K. Tanaka, S. Viti, Y. T. Yan, **W. Yang**, R. Q. Mao, D. H. Quan, 2025, *A&A*, 696, A31

14. [Molecular inventory of a young eruptive star's environment Case study of the classical FU Orionis star V1057 Cyg](#)

Zs. M. Szabó, A. Belloche, K. M. Menten, Y. Gong, Á. Kóspál, P. Ábrahám, **W. Yang**, C. J. Cyganowski, F. Wyrowski, 2025, *A&A*, 694, A392

13. [Hyperfine structure of methanol molecule as traced by Class I methanol masers](#)

I. I. Agafonova, O. S. Bayandina, Y. Gong, C. Henkel, Kee-Tae Kim, M. G. Kozlov, B. Lankhaar, S. A. Levshakov, K. M. Menten, W. Ubachs, I. E. Val'tts, **W. Yang** (alphabetical), 2024, *MNRAS*, 533, 1714

12. [First detection of the \$J_{-1}-\(J-1\)_0-E\$ methanol maser transitions at \$J=7\$ and 10](#)

Pedro K. Humire, Gisela Ortiz-León, Antonio Hernández-Gómez, **Wenjin Yang**, Christian Henkel, Sergio Martín, 2024, *A&A*, 688, L1

11. [Discovery of widespread non-metastable ammonia masers in the Milky Way](#)

Y. T. Yan, C. Henkel, K. M. Menten, T. L. Wilson, A. Wootten, Y. Gong, F. Wyrowski, **W. Yang**, A. Brunthaler, A. Kraus, B. Winkel, 2024, *A&A*, 686, A205

10. Molecular Bubble and Outflow in S Mon Revealed by Multi-band Datasets

Dejian Liu, Ye Xu, Yingjie Li, Zehao Lin, Chaojie Hao, **Wenjin Yang**, Jingjing Li, Xinrong Liu, Yiwei Dong, Shuaibo Bian, Deyun Kong, 2024, *ApJ*, 964, 93

9. Sulfur Isotope Ratios in the Large Magellanic Cloud

Y. Gong, C. Henkel, K. M. Menten, C.-H. R. Chen, Z. Y. Zhang, Y. T. Yan, A. Weiss, N. Langer, J. Z. Wang, R. Q. Mao, X. D. Tang, **W. Yang**, Y. P. Ao, M. Wang, 2023, *A&A*, 679, L6

8. Protonated hydrogen cyanide as a tracer of pristine molecular gas

Y. Gong, F. J. Du, C. Henkel, A. M. Jacob, A. Belloche, J. Z. Wang, K. M. Menten, **W. Yang**, D. H. Quan, C. T. Bop, G. N. Ortiz-León, X. D. Tang, M. R. Rugel, S. Liu, 2023, *A&A*, 679, A39

7. The Effelsberg survey of FU Orionis and EX Lupi objects II. H₂O maser observations

Zs. M. Szabó, Y. Gong, **W. Yang**, K. M. Menten, O. S. Bayandina, C. J. Cyganowski, Á. Kóspál, P. Ábrahám, A. Belloche, F. Wyrowski, 2023, *A&A*, 674, A202

6. The Effelsberg survey of FU Orionis and EX Lupi objects. I. Host environments of FUors and EXors traced by NH₃

Zs. M. Szabó, Y. Gong, K. M. Menten, **W. Yang**, C. J. Cyganowski, Á. Kóspál, P. Ábrahám, A. Belloche, F. Wyrowski, 2023, *A&A*, 672, A158

5. Widespread subsonic turbulence in Ophiuchus North 1

Yan Gong, Shu Liu, Junzhi Wang, Weishan Zhu, Guang-Xing Li, **Wenjin Yang**, Jixian Sun, 2022, *A&A*, 663, A82

4. Light Deflection under the Gravitational Field of Jupiter-Testing General Relativity

Yingjie Li, Ye Xu, Jingjing Li, Yuanwei Wu, Shaibo Bian, Zehao Lin, **Wenjin Yang**, Chaojie Hao, Dejian Liu, 2022, *ApJ*, 925, 47

3. Probing the electron-to-proton mass ratio gradient in the Milky Way with Class I methanol masers

S. A. Levshakov, I. I. Agafonova, C. Henkel, Kee-Tae Kim, M. G. Kozlov, B. Lankhaar, **W. Yang**, 2022, *MNRAS*, 511, 413

2. Searching for further evidence for cloud-cloud collisions in L1188

Y. Gong, X. D. Tang, C. Henkel, K. M. Menten, R. Q. Mao, Y. Wang, M.-Y. Lee, W. S. Zhu, Y. Lin, S. B. Zhang, X. P. Chen, **W. J. Yang**, 2019, *A&A*, 632, A115

1. Molecular Gas toward the Gemini OB1 Molecular Cloud Complex. II. CO Outflow Candidates with Possible WISE Associations

Yingjie Li, Fa-Cheng Li, Ye Xu, Chen Wang, Xin-Yu Du, **Wenjin Yang**, Ji Yang, 2018, *ApJS*, 235, 15

PROCEEDINGS

4. The molecular inventory of a young eruptive star's environment: Case study of the classical FU Orionis star, V1057 Cyg

Zsófia Marianna Szabó, Arnaud Belloche, Karl M. Menten, Yan Gong, **Wenjin Yang**, Ágnes Kóspál, Péter Ábrahám, Friedrich Wyrowski, Claudia J. Cyganowski, 2024, *EAS*, 398

3. ATLASGAL: methanol masers at 3 mm

W. Yang, Y. Gong, K. M. Menten, F. Wyrowski, J. S. Urquhart, C. Henkel, T. Csengeri, S. P. Ellingsen, A. R. Bemis, J. Jang, 2024, *IAU*, 380, 266

2. H₂O masers and host environments of FU Orionis and EX Lupi type low-mass eruptive YSOs

Zsófia Marianna Szabó, Yan Gong, **Wenjin Yang**, Karl M. Menten, Olga S. Bayandina, Claudia J. Cyganowski, Ágnes Kóspál, Péter Ábrahám, Arnaud Belloche, Friedrich Wyrowski, 2024, *IAU*, 380, 246

1. Searching masers from the Sagittarius stellar stream

Yuanwei Wu, Bo Zhang, Yan Gong, **Wenjin Yang**, Nicolas Maun, 2024, *IAU*, 380, 128